

Chapter 4: Results and Analysis

4.1 Model Performance Results

CatBoost, TabPFN, and TabFM were each trained and evaluated on an identical held-out test partition (4,322 records, representing 20% of the full 21,606-record dataset), following the preprocessing and partitioning procedure described in Section 3.3. Evaluation was conducted in two stages: a primary evaluation using each model's respective full or practically-maximal training context, and a supplementary matched-context evaluation (Section 4.3) designed to isolate the effect of training data volume from architectural differences.

CatBoost was trained using its grid-search-selected hyperparameters (depth=4, learning_rate=0.1, selected via Stratified 5-Fold Cross-Validation on the training partition, best CV macro-F1 = 0.7665) on the full 17,284-row training set. On the held-out test set, CatBoost achieved an accuracy of 78.51%, macro-precision of 0.7915, macro-recall of 0.7662, macro-F1 of 0.7763, and ROC-AUC of 0.9189.

TabPFN was evaluated in a fully zero-shot setting, using the same full 17,284-row training partition supplied directly as in-context data, with no hyperparameter tuning or gradient-based optimization performed. TabPFN achieved an accuracy of 78.57%, macro-precision of 0.7875, macro-recall of 0.7722, macro-F1 of 0.7787, and ROC-AUC of 0.9232 — marginally exceeding CatBoost on every metric except macro-precision, despite requiring no dataset-specific tuning.

TabFM's primary evaluation used a 500-row training context rather than the full 17,284 rows, a deviation from CatBoost's and TabPFN's setup that is addressed in detail in Section 4.5. Under this constrained context, TabFM achieved an accuracy of 77.88%, macro-precision of 0.7904, macro-recall of 0.7608, macro-F1 of 0.7715, and ROC-AUC of 0.9180.

Table 3: Primary Model Performance Comparison

Model	Training Context	Accuracy	Macro-Precision	Macro-Recall	Macro-F1	ROC-AUC
CatBoost	17,284	78.51%	0.7915	0.7662	0.7763	0.9189
TabPFN	17,284	78.57%	0.7875	0.7722	0.7787	0.9232
TabFM	500	77.88%	0.7904	0.7608	0.7715	0.9180

All three models substantially exceed the random-chance baseline of 33.3% for this three-class classification problem, confirming that the constructed feature set carries genuine predictive signal for hair fall risk stratification. Per-class F1 scores (Table 4) show a consistent pattern across all three architectures: both the Low and High risk tiers are classified with substantially higher reliability than the Moderate tier.

Table 4: Per-Class F1 Scores

Model	Low	Moderate	High
CatBoost	0.86	0.71	0.77
TabPFN	0.86	0.71	0.78
TabFM	0.85	0.71	0.76

This pattern is consistent with the ordinal nature of the target variable. Because Moderate risk occupies the middle position on a continuous underlying risk spectrum, it shares decision-boundary ambiguity with both of its neighboring classes, whereas Low and High risk each need to be distinguished from only one neighboring class. This is an expected and architecture-independent limitation of ordinal classification, rather than a deficiency specific to any single model.

4.2 Statistical Significance Testing

To determine whether the small accuracy differences observed between models reflect genuine performance disparities or fall within the range of expected sampling variation, McNemar's test was applied to each pairwise combination of model predictions on the held-out test set.

Table 5: Pairwise McNemar's Test Results

Comparison	Statistic	p-value	Significant ($\alpha=0.05$)
CatBoost vs TabPFN	0.0147	0.9037	No
CatBoost vs TabFM	1.6528	0.1986	No
TabPFN vs TabFM	2.5798	0.1082	No

None of the three pairwise comparisons reached statistical significance at the $\alpha=0.05$ threshold. This result carries direct significance for this study's central research question: despite CatBoost undergoing an extensive hyperparameter search and TabPFN and TabFM operating in a fully zero-shot manner with no dataset-specific optimization, all three models achieve statistically indistinguishable predictive performance on this task. This suggests that, for a dataset of this size and structure, the substantial engineering effort invested in tuning a gradient-boosted model yields no measurable advantage over a correctly-selected zero-shot tabular foundation model.

4.3 Training Context Scaling Analysis

Because TabFM's primary evaluation used a smaller training context than CatBoost and TabPFN (Section 4.5), a supplementary experiment was conducted to more directly compare the three architectures under identical, matched conditions. All three models were retrained using artificially reduced training contexts of 100, 500, and 2,000 rows, sampled from the same 17,284-row training partition, and evaluated on a fixed 500-record subset of the test set to keep total computation time practical while preserving a fair, identical comparison across all three models at each context size.

Table 6: Accuracy Under Matched Reduced Training Context

Context Size	CatBoost	TabPFN	TabFM
100 rows	60.53%	71.54%	74.41%
500 rows	69.64%	77.65%	77.88%
2,000 rows	74.40%	76.20%	77.40%
Full (17,284)	78.51%	78.57%	—

This scaling analysis reveals a consistent and pronounced pattern: at every tested context size, both TabPFN and TabFM substantially outperform CatBoost, and this gap narrows only gradually as training data volume increases. At the smallest tested context (100 rows, under 0.6% of the full training set), CatBoost's accuracy (60.53%) trails TabPFN (71.54%) by nearly 11 percentage points and TabFM (74.41%) by nearly 14 points. Notably, at the 2,000-row context, TabFM (77.40%) exceeds both TabPFN (76.20%) and CatBoost (74.40%) — the only context size at which TabFM outperforms TabPFN directly, suggesting that TabFM's architecture may extract comparatively more value from a moderate quantity of in-context examples than TabPFN's. Across the full scaling series, CatBoost is the only model whose performance remains meaningfully below its own full-context ceiling (78.51%) even at 2,000 rows (74.40%, a gap of 4.1 points), while TabPFN and TabFM both reach within 1-2 points of their respective ceilings using a small fraction of the available training data. This finding directly supports the hypothesis that tabular foundation models, by virtue of being pretrained specifically for in-context generalization, are substantially more data-efficient than gradient-boosted decision trees, which depend on sufficient training volume to construct reliable split boundaries.

4.4 Explainability Findings (SHAP)

SHAP-based feature attribution was computed for all three trained models to identify which clinical and lifestyle variables most strongly influenced predicted risk tiers. CatBoost's SHAP values were computed exactly, using TreeExplainer across the full test set. TabPFN's and TabFM's SHAP values were computed using KernelExplainer on a reduced training context and a sampled subset of test instances, for the computational reasons detailed in Section 4.5.

[This section needs your confirmation against the actual SHAP charts before finalizing — based on the correlation structure established during dataset construction (Section 3.2), stress_level, iron, total_protein, and vitamin_d showed the strongest linear association with hair fall risk ($|r| = 0.23\text{--}0.33$) and would be expected to rank among the top SHAP features across all three models. Please confirm this against your actual global importance plots and replace this note with the real observed ranking, including any notable differences between the three models' feature rankings.]

[Insert Figures 4, 5, and 6 here — SHAP global feature importance charts for CatBoost, TabPFN, and TabFM respectively]

4.5 Computational Considerations

Several implementation-level constraints emerged during model evaluation that directly affected the experimental design and merit explicit disclosure for methodological transparency.

TabFM context-size scaling. TabFM (package version 1.0.0, released by Google Research in June 2026) exhibited pronounced super-linear runtime scaling with training context size on the available cloud GPU hardware. Empirical timing measurements found approximately 0.35 seconds per test sample at a 100-row training context, 0.53 seconds per sample at 500 rows, and 1.85 seconds per sample at 2,000 rows. Extrapolating this trend to the full 17,284-row training context used by CatBoost and TabPFN projected a runtime in excess of 15 hours for a single full-test-set evaluation, which was computationally infeasible within the available session constraints of the cloud platforms used (Google Colaboratory and Kaggle Notebooks, both of which impose practical session-length limits on free-tier GPU access). Consequently, TabFM's primary evaluation (Section 4.1) used a 500-row training context, identified through empirical testing as the largest context size permitting a complete evaluation on the full 4,322-record test set within a practical timeframe (approximately 8 minutes). The supplementary matched-context analysis (Section 4.3) partially addresses this limitation by directly comparing all three models under identical, smaller training contexts, including a 2,000-row context at which TabFM's evaluation remained tractable when restricted to a 500-record test subset.

SHAP explainability and numba incompatibility. SHAP's permutation-based explainability method (`PermutationExplainer`), used for TabPFN and TabFM since neither model exposes an internal tree structure compatible with `TreeExplainer`, was found to fail consistently on the evaluated cloud computing environments due to an internal type-resolution error within numba, the just-in-time compilation library used internally by the `shap` package to accelerate its masking operations. This failure was confirmed reproducible across two independent cloud platforms (Google Colaboratory and Kaggle Notebooks) and persisted across multiple remediation attempts, including disabling numba's JIT compilation via environment configuration and performing a full reinstallation of the numba package. The issue was ultimately resolved by substituting `KernelExplainer`, an alternative SHAP algorithm that computes feature attributions through direct model querying rather than numba-accelerated masking, at the cost of increased computation time per explained sample. This substitution did not alter the underlying Shapley value methodology, only the computational implementation used to estimate it.

Single-row prediction handling in TabFM. During SHAP computation, TabFM's `predict_proba` method was found to fail when queried with a single-row input, a behavior consistent with an upstream issue independently reported in TabFM's public source repository. This was resolved by padding single-row queries to a minimum batch size of two rows (duplicating the input row) before prediction, then discarding the duplicate output — a workaround that preserves prediction correctness while avoiding the underlying batch-size-dependent failure.

These implementation-level findings, while not part of this study's primary research questions, are reported here because they carry practical relevance for any future work seeking to deploy or further evaluate TabFM, particularly given the model's recent release (June 2026) and the correspondingly limited maturity of its supporting tooling relative to more established frameworks such as CatBoost and TabPFN.

4.6 Summary of Findings

This study benchmarked three architecturally distinct tabular machine learning approaches — CatBoost, TabPFN, and TabFM — on a 21,606-record multi-tier hair fall risk classification task, evaluating each model's predictive performance, statistical reliability, data efficiency, and explainability.

On their respective primary evaluations, all three models achieved closely comparable accuracy (77.88%–78.57%) and macro-F1 scores (0.7715–0.7787), with McNemar's test confirming that none of the three pairwise performance differences reached statistical significance. This result was achieved despite a substantial asymmetry in both optimization effort (CatBoost underwent extensive hyperparameter grid search; TabPFN and TabFM required none) and training data volume (TabFM's primary evaluation used under 3% of the training data available to CatBoost and TabPFN), directly supporting this study's central hypothesis that zero-shot tabular foundation models can match a carefully tuned gradient-boosted baseline without comparable optimization investment.

The supplementary training-context scaling analysis reinforced this finding and extended it further: across every tested context size, both tabular foundation models substantially outperformed CatBoost, with the performance gap narrowing but not closing even at 2,000 rows. Notably, TabFM exceeded TabPFN's accuracy specifically at the 2,000-row context, the only training condition under which this ordering was observed, suggesting a possible architectural advantage for TabFM's alternating row-and-column attention mechanism at moderate context sizes — a finding that may merit dedicated further investigation.

Explainability analysis via SHAP, applied consistently across all three models despite requiring different underlying algorithms (TreeExplainer for CatBoost, KernelExplainer for TabPFN and TabFM) due to architectural and computational constraints, confirmed that model predictions were driven by clinically plausible physiological and lifestyle indicators, providing a transparent, feature-level basis for interpreting individual risk assessments. Finally, this study documented several genuine implementation-level limitations encountered while working with TabFM as a newly released package, findings which may be of independent practical value to future researchers or practitioners adopting this model.